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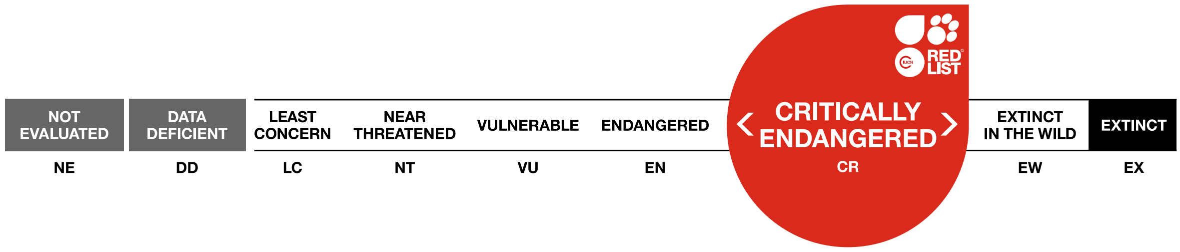
Pongo abelii, Sumatran Orangutan

Amendment version

Assessment by: Singleton, I., Wich , S.A., Nowak, M., Usher, G. & Utami-Atmoko, S.S.



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Taxonomy

Kingdom	Phylum	Class	Order	Family
Animalia	Chordata	Mammalia	Primates	Hominidae

Scientific Name: *Pongo abelii* Lesson, 1827

Common Name(s):

- English: Sumatran Orangutan
- French: Orang-outan de Sumatra
- Spanish; Castilian: Orangután de Sumatra
- German: Sumatra-Orang-Utan

Taxonomic Source(s):

Nater, A., Greminger, M.P., Nurcahyo, A., Nowak, M.G., de Manuel Montero, M., Desai, T., Groves, C.P., Pybus, M., Sonay, T.B., Roos, C., Lameira, A.R., Wich, S.A., Askew, J., Davila-Ross, M., Fredriksson, G.M., de Valles, G., Casals, F., Prado-Martinez, J., Goossens, B., Verschoor, E.J., Warren, K.S., Singleton, I., Marques, D.A., Pamungkas, J., Perwitasari-Farajallah, D., Rianti, P., Tuuga, A., Gut, I.G., Gut, M., Orozco-terWengel, P., van Schaik, C.P., Bertranpetit, J., Anisimova, M., Scally, A., Marques-Bonet, T., Meijaard, E. and Krützen, M. 2017. Morphometric, behavioural, and genomic evidence for a new orangutan species. *Current Biology* 27: DOI: 10.1016/j.cub.2017.09.047.

Taxonomic Notes:

Historically, the Sumatran and Bornean Orangutans were both considered to be subspecies of *Pongo pygmaeus* (e.g., Rijksen and Meijaard 1999). Recent taxonomic reviews (Groves 2001, Brandon-Jones *et al*. 2004) support the acceptance of the Sumatran Orangutan (*Pongo abelii*) as distinct from its Bornean relative (*Pongo pygmaeus*). This classification has since been widely adopted (e.g., Singleton *et al*. 2004). A recent study (Nater *et al*. 2017) proposed that the Batang Toru population is a separate species (*Pongo tapanuliensis*), and therefore this assessment does not include the orangutans in that area.

Assessment Information

Red List Category & Criteria: Critically Endangered A4bcd ver 3.1

Year Published: 2024

Date Assessed: October 16, 2017

Justification:

Due to high levels of habitat conversion and fragmentation, and illegal killing, *Pongo abelii* is estimated to have experienced a significant population reduction in recent years. Forest loss data indicate that key Sumatran Orangutan forest habitat (i.e., below 500 m asl) was reduced by 60% of its area between 1985 and 2007 (Wich *et al*. 2008, 2011). It is thought that this reduction will continue as forests within the species' range remain under considerable threat (Wich *et al*. 2016). When relative stability returned to Aceh in 2005 after several years of civil conflict, pressure on natural habitats increased dramatically (Wich *et al*. 2011). Significant areas of the Orangutan's range are seriously threatened by logging, mining concessions and agricultural plantations, while new roads are continuously being cut through the habitat. Even in formally protected areas, Orangutans remain under threat from forest conversion to plantations, illegal settlement and encroachment (Wich *et al*. 2008, 2011, 2016). Furthermore, an illegal spatial land-use plan being implemented by the Government of Aceh Province ignores the Leuser Ecosystem's status as a 'National Strategic Area', designated for its environmental function. Moreover, modelling based on different land-use scenarios and their likely impacts predicts that approximately an additional 4,000 Sumatran Orangutans could be lost by 2030 as a direct consequence of this spatial plan and related developments, and that in 2060 there could be a 81% decline of the population compared to the population in 1985 (Wich *et al.* 2016, Nowak unpublished data).

Due to their slow life history, with a generation time of at least 25 years (Wich *et al*. 2004, 2009), Sumatran Orangutan populations are unable to sustain substantial and continual loss of individuals. If the rate of decline observed since 1985 and predicted continues unabated, the population decline will exceed 80% over a three-generation period (i.e., 75 years from 1985 to 2060), hence qualifying *Pongo abelii* as Critically Endangered under criterion A4.

Previously Published Red List Assessments

2023 – Critically Endangered (CR)

2017 – Critically Endangered (CR)

Geographic Range

Range Description:

Pongo abelii is endemic to the island of Sumatra, Indonesia. It is restricted to the north of the island, with its southern limit being the Simpang Kanan River and tributaries on the west coast and the Asahan River on the east coast, and its northern limit coinciding primarily with the northern boundary of the Leuser Ecosystem in Aceh Province (Wich *et al*. 2003, 2008, 2016).

Today the majority of Sumatran Orangutans (82.5%) are found in Aceh Province at the northernmost tip of the island. There are populations in North Sumatra Province but the largest of these, in the southern and eastern regions of the Leuser Ecosystem, straddles the border with Aceh. Despite a few smaller patches of forest south of the Leuser Ecosystem appearing to still host orangutan populations, only one entirely North Sumatran population is considered viable in the long term, namely the Pakpak Barat population (for precise locations, see Wich *et al.* 2008, 2016).

Sumatran Orangutan densities decline with increasing altitude and few, if any, reproducing populations are thought to reside in forests above 1,500 m asl (Wich *et al*. 2016).

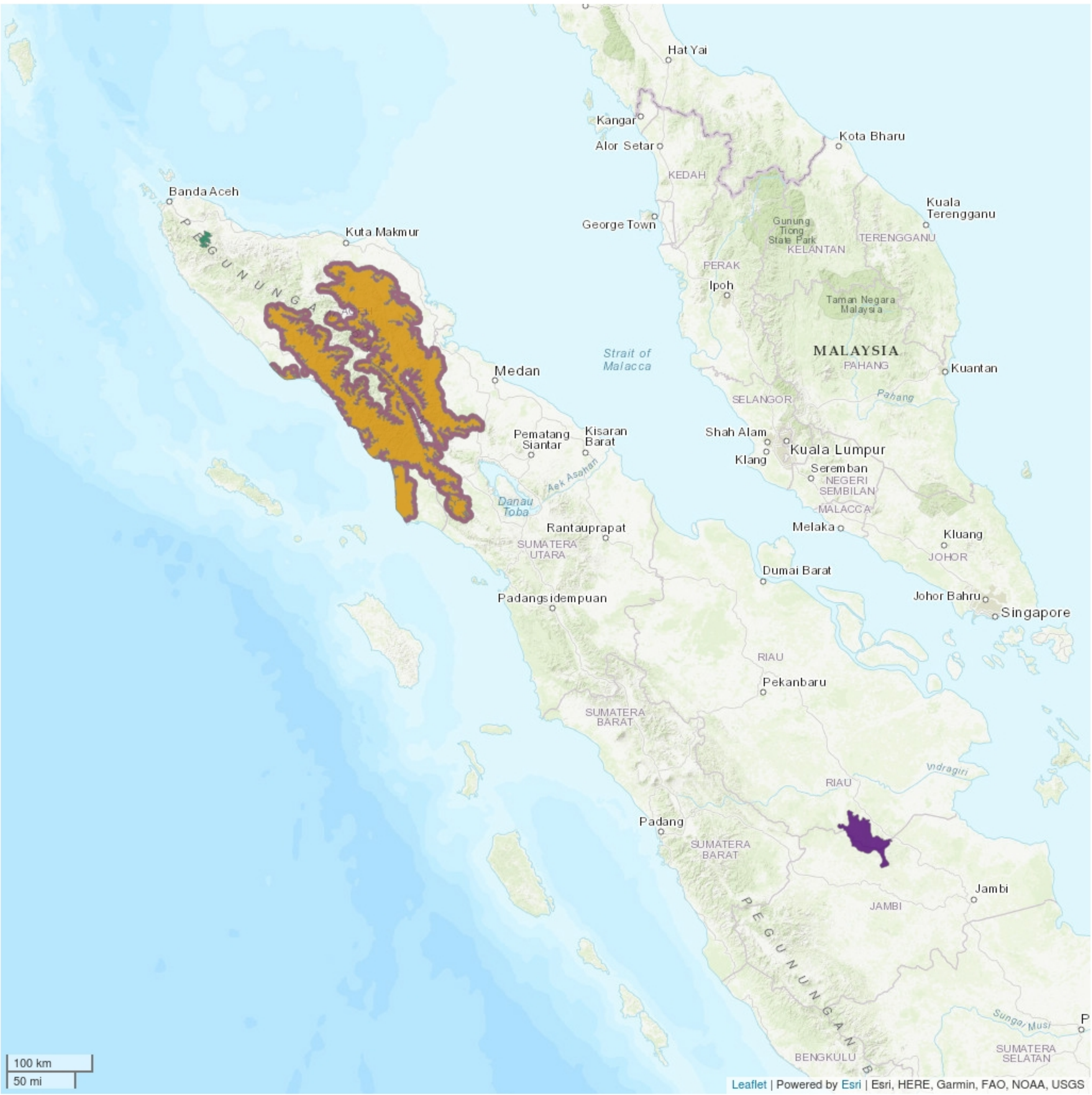
See the Supplementary Information for details of how the distribution map for this species was updated for the amended version of this assessment.

For further information about this species, see Supplementary Material.

Country Occurrence:

Native, Extant (resident): Indonesia (Sumatera)

Distribution Map

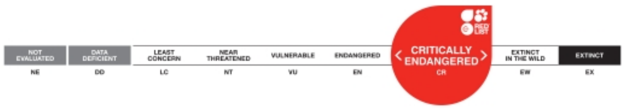


Legend

- EXTANT (RESIDENT)
- POSSIBLY EXTANT (RESIDENT)
- EXTANT & REINTRODUCED (RESIDENT)
- EXTANT & INTRODUCED (RESIDENT)

Compiled by:

Wich, S.A., Ni'Matullah, S, Sherman, J.,and Meijaard, E. 2023. Spatial data for P. abelii Red Listing. Borneo Futures, Brunei Darussalam 2023



The boundaries and names shown and the designations used on this map do not imply any official endorsement, acceptance or opinion by IUCN.

Population

The most recent population estimate for the Sumatran Orangutan is 13,846 individuals, in a total area of 16,775 km² of forest (Wich *et al.* 2016). Excluding populations of fewer than 250 individuals (i.e., considering only populations that are potentially viable over the long term) leaves just 13,587 individuals. The vast majority (i.e., 95.0%) occur in the Leuser Ecosystem, while other populations are found in the Sidiangkat and Pakpak. The 2016 estimate is higher than the previous estimate of around 6,600 individuals remaining (Wich *et al*. 2008), as it takes into account three factors: a) orangutans were found in greater numbers at higher altitudes than previously supposed (i.e., up to 1,500 m asl not just to 1,000 m asl), b) they were found to be more widely distributed in selectively-logged forests than previously assumed, and c) orangutans were found in some previously unsurveyed forest patches. The new estimate does not, therefore, reflect a real increase in Sumatran Orangutan numbers. On the contrary, it reflects only much improved survey techniques and coverage, and hence more accurate data. It is extremely important to note, therefore, that overall numbers continue to decline dramatically.

Since Sumatran Orangutans have been found up to 1,500 m asl in many areas, there is probably better connectivity among subpopulations in the mountainous Leuser Ecosystem than was previously considered the case (Wich *et al*. 2008). Nevertheless, recent studies found genetic differentiation between subpopulations that is at least partially due to geographic barriers, such as rivers and high mountain ridges, even within the Leuser Ecosystem (Nater *et al*. 2013).

In addition to the wild populations, two entirely new Sumatran Orangutan populations are gradually being established via the reintroduction of confiscated illegal pets; one in and around the Bukit Tigapuluh National Park (Jambi and Riau provinces) and one in and around the Jantho Pine Forest Nature Reserve, in the far north of Aceh. To date, more than 260 individuals have been reintroduced. The goal of these efforts is to eventually establish new, genetically-viable, fully-reproducing and self-sustaining wild populations as a safety net against catastrophe elsewhere in the species' range.

About 35.6% of the Sumatran Orangutan population is in protected areas (World Database Protected Areas recognized areas, i.e. excluding hutan lindung/protection forest; SOCP unpublished data), followed by 40.0% in hutan lindung/protection forest, 3.6% in limited production forest, 11.8% in production forest, and 8.9% in other use forest. Including the whole Leuser Ecosystem, which is on the list of Indonesia's WDPAs: 98.5% of all Sumatran Orangutans are in PAs (incl. hutan lindung/protection forest) or 95.5% in all WDPA recognized areas (thus excluding hutan lindung/protection forest).
Current Population Trend: Decreasing

Habitat and Ecology **(see Appendix for additional information)**

Sumatran Orangutans inhabit moist lowland forest, montane forest and peat swamps. They are diurnal and almost exclusively arboreal; females virtually never travel on the ground and adult males do so only rarely. This contrasts somewhat with Bornean Orangutans (especially adult males), which descend and travel on the ground more often (Ancrenaz *et al*. 2014). While both species depend on high-quality primary forests, Bornean Orangutans appear better able to tolerate limited habitat disturbance (Husson *et al*. 2009). In Sumatra, densities plummet by up to 60%, even if logging is highly selective (see Rao and van Schaik 1997) and Orangutan behavioural ecology differs in primary and logged forests (Hardus *et al*. 2012a).

Sumatran Orangutans are primarily frugivores, but also eat young and mature leaves, seeds, shoots, pith, flowers, insects (termites and ants), bark and, on occasion, the meat of slow loris (Fox *et al*. 2004, Wich *et al*. 2006, Morrogh-Bernard *et al*. 2009, Hardus *et al*. 2012b). Female home ranges vary in size from 0.8–1.5 km². The true extent of male home range size is not fully known, although ranges well in excess of 30 km² are inferred (Singleton and van Schaik 2001, Singleton *et al*. 2009).

Females first give birth at about 15 years of age (Wich *et al*. 2004, 2009). Sumatran Orangutans have the longest interbirth intervals of any mammalian species, ranging from 8.2 to 9.3 years (compared with 6.1 to 7.7 years for *P. pygmaeus*; Wich *et al*. 2004, 2009, van Noordwijk and van Schaik 2005). Gestation lasts approximately 254 days (Kingsley 1981). Infants are weaned at *ca* 7 years. On average, a female bears 4–5 offspring during her lifetime. Males mature at about about 14 years (Wich *et al*. 2004). They exhibit bimaturism, whereby fully flanged adult males and smaller unflanged males are both capable of reproducing, but employ differing mating strategies (Utami *et al*. 2002, Utami Atmoko *et al*. 2009). Longevity in the wild has been estimated at 53 years for females and 58 years for males (Wich *et al*. 2004). Generation length is at least 25 years (Wich *et* al. 2004, 2009).

Systems: Terrestrial

Use and Trade

It is illegal to capture, injure, kill, own, keep, transport, or trade a Sumatran Orangutan.

Threats (see Appendix for additional information)

The Sumatran Orangutan's survival is seriously threatened by habitat loss and fragmentation (Wich *et al*. 2008, 2011, 2016). Forests continue to be cleared at the large and medium scale for oil-palm plantations that can each cover hundreds of square kilometres. On a smaller scale, logging for timber (both legal and illegal) remains a threat, as does the creation of new roads, which fragment populations and gives access to illegal settlements and further encroachment for agriculture and plantations (also frequently illegal), and to wildlife poachers. When industrial plantations are established, the resident orangutans are forced to seek refuge in adjacent forest patches, if any remain, but in the long term they are likely to succumb to malnutrition and starvation due to competition and limited resources. Such forest fragments are often subsequently cleared as well.
br/Sumatran Orangutans are frequently killed deliberately, completely illegally, and surviving infants end up in an illegal pet trade. This trade tends to be a by-product of habitat conversion, for example, if an Orangutan is found in an isolated patch of trees during the conversion process, there is a high probability it will be killed. Sumatran Orangutans are also regularly killed in human-wildlife conflict situations, for example, if raiding fruit crops on farmland at the forest edge (Wich *et al.* 2012).

By far the largest single current threat to the Sumatran Orangutan comes from a spatial land-use plan ratified by the government of Aceh Province in 2013. Conservation of the Leuser Ecosystem being obligated by Aceh’s own special autonomy law (Law No 11 2006) due to its designation in 2007/8 as a National Strategic Area for its environmental function (which requires its inclusion and special consideration at all levels of spatial planning). But alarmingly the current Aceh spatial plan completely ignores the Leuser Ecosystem’s existence. Both the provincial and national governments have publicly acknowledged the illegality of the Aceh spatial plan, but to date it remains officially ratified at the provincial level and has not been formally cancelled by the national government. As of early 2016, legal challenges are attempting to rectify this situation and to have the current Aceh spatial plan replaced with one that complies fully with all relevant laws, is based on sound environmental sensitivity analyses, and affords appropriate protection to the Leuser Ecosystem in accordance with existing legislation. However, whilst the status quo persists, the Aceh Provincial Spatial Plan of 2013 allows huge tracts of Sumatran Orangutan habitat to be designated for new plantations, and timber and mining concessions, and will lead to many more Sumatran Orangutans being lost in the ensuing years (Wich *et al*. 2016). The existing plan also effectively legitimizes numerous roads cut illegally through the forest, which further fragment Orangutan populations and provide yet more access to hunting and encroachment.

Conservation Actions (see Appendix for additional information)

Pongo abelii is strictly protected by Indonesian National Law No. 5/1990 on the Conservation of Natural Resources and Ecosystems, under which it is illegal to capture, injure, kill, own, keep, transport, or trade a Sumatran Orangutan. Sumatran Orangutans are also protected by international legislation, and listed on CITES Appendix I.

Protection of large areas of primary forest below 1,500 m asl is needed to secure their long-term future. The species' major stronghold is the Leuser Ecosystem – an area of 26,000 km² mostly contiguous forest that supports *circa* 95.0% of the Sumatran Orangutans remaining in the wild. Conservation of the Leuser Ecosystem was called for under Indonesian National Law No. 11/2006 concerning Governance in Aceh, and it was inaugurated by Presidential Decree in 1998. Designated a National Strategic Area for its environmental function, the Leuser Ecosystem must be fully recognized and its integrity ensured at all levels of spatial land-use planning. According to these laws, management of the Leuser Ecosystem does not exclude non-forest uses, but stresses the importance of sustainable management with conservation of natural resources as the primary goal.

Within the Leuser Ecosystem is the 9,000 km² Gunung Leuser National Park, also designated a Man and Biosphere Reserve and a part of the Tropical Rainforest Heritage of Sumatra World Heritage Cluster Site by UNESCO. The park alone, comprising mostly high mountains, supports only 25.3% of Sumatra's Orangutans. Much of Sumatra's dense lowland forest is outside the National Park's boundaries, but is part of the larger Leuser Ecosystem. Also within the Leuser Ecosystem, but not a part of the World Heritage Cluster Site, is the 1,025 km² Singkil Swamps Wildlife Reserve. Outside the Leuser Ecosystem, no other large, formally-established conservation areas harbour this species.

Credits

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External Resources

For Supplementary Material, and for Images and External Links to Additional Information, please see the Red List website.

Appendix

Habitats

(http://www.iucnredlist.org/technical-documents/classification-schemes)

Habitat	Season	Suitability	Major Importance?
1. Forest -> 1.6. Forest - Subtropical/Tropical Moist Lowland	Resident	Suitable	No
1. Forest -> 1.8. Forest - Subtropical/Tropical Swamp	Resident	Suitable	Yes
1. Forest -> 1.9. Forest - Subtropical/Tropical Moist Montane	Resident	Suitable	Yes

Use and Trade

(http://www.iucnredlist.org/technical-documents/classification-schemes)

End Use	Local	National	International
13. Pets/display animals, horticulture	No	No	Yes

Threats

(http://www.iucnredlist.org/technical-documents/classification-schemes)

Threat	Timing	Scope	Severity
1. Residential & commercial development -> 1.1. Housing & urban areas	Ongoing	Minority (<50%)	Slow, significant declines
Stresses:	1. Ecosystem stresses -> 1.1. Ecosystem conversion 1. Ecosystem stresses -> 1.2. Ecosystem degradation		
1. Residential & commercial development -> 1.3. Tourism & recreation areas	Ongoing	Minority (<50%)	Slow, significant declines
Stresses:	1. Ecosystem stresses -> 1.1. Ecosystem conversion 2. Species Stresses -> 2.2. Species disturbance		
2. Agriculture & aquaculture -> 2.1. Annual & perennial non-timber crops -> 2.1.2. Small-holder farming	Ongoing	Minority (<50%)	Rapid declines
Stresses:	1. Ecosystem stresses -> 1.1. Ecosystem conversion 1. Ecosystem stresses -> 1.2. Ecosystem degradation		
2. Agriculture & aquaculture -> 2.1. Annual & perennial non-timber crops -> 2.1.3. Agro-industry farming	Ongoing	Majority (50-90%)	Very rapid declines
Stresses:	1. Ecosystem stresses -> 1.1. Ecosystem conversion 1. Ecosystem stresses -> 1.2. Ecosystem degradation		
3. Energy production & mining -> 3.2. Mining & quarrying	Ongoing	Minority (<50%)	Slow, significant declines
Stresses:	1. Ecosystem stresses -> 1.1. Ecosystem conversion 1. Ecosystem stresses -> 1.2. Ecosystem degradation 2. Species Stresses -> 2.2. Species disturbance		
4. Transportation & service corridors -> 4.1. Roads & railroads	Ongoing	Majority (50-90%)	Very rapid declines

https://dx.doi.org/10.2305/IUCN.UK.2024-1.RLTS.T121097935A259045437.en

Stresses:	1. Ecosystem stresses -> 1.1. Ecosystem conversion 1. Ecosystem stresses -> 1.2. Ecosystem degradation		
5. Biological resource use -> 5.1. Hunting & trapping terrestrial animals -> 5.1.1. Intentional use (species is the target)	Ongoing	Minority (<50%)	Very rapid declines
Stresses:	2. Species Stresses -> 2.1. Species mortality		
5. Biological resource use -> 5.1. Hunting & trapping terrestrial animals -> 5.1.2. Unintentional effects (species is not the target)	Ongoing	Minority (<50%)	Rapid declines
Stresses:	2. Species Stresses -> 2.1. Species mortality		
5. Biological resource use -> 5.1. Hunting & trapping terrestrial animals -> 5.1.3. Persecution/control	Ongoing	Minority (<50%)	Rapid declines
Stresses:	2. Species Stresses -> 2.1. Species mortality		
5. Biological resource use -> 5.3. Logging & wood harvesting -> 5.3.3. Unintentional effects: (subsistence/small scale) [harvest]	Ongoing	Minority (<50%)	Slow, significant declines
Stresses:	1. Ecosystem stresses -> 1.1. Ecosystem conversion 1. Ecosystem stresses -> 1.2. Ecosystem degradation		
5. Biological resource use -> 5.3. Logging & wood harvesting -> 5.3.4. Unintentional effects: (large scale) [harvest]	Ongoing	Minority (<50%)	Slow, significant declines
Stresses:	1. Ecosystem stresses -> 1.1. Ecosystem conversion 1. Ecosystem stresses -> 1.2. Ecosystem degradation 2. Species Stresses -> 2.1. Species mortality 2. Species Stresses -> 2.2. Species disturbance		
7. Natural system modifications -> 7.1. Fire & fire suppression -> 7.1.1. Increase in fire frequency/intensity	Ongoing	Majority (50-90%)	Very rapid declines
Stresses:	1. Ecosystem stresses -> 1.1. Ecosystem conversion 2. Species Stresses -> 2.1. Species mortality 2. Species Stresses -> 2.2. Species disturbance		
11. Climate change & severe weather -> 11.1. Habitat shifting & alteration	Ongoing	Whole (>90%)	Slow, significant declines
Stresses:	1. Ecosystem stresses -> 1.3. Indirect ecosystem effects 2. Species Stresses -> 2.3. Indirect species effects -> 2.3.7. Reduced reproductive success		

Conservation Actions in Place
(http://www.iucnredlist.org/technical-documents/classification-schemes)

Conservation Action in Place
In-place research and monitoring
Action Recovery Plan: Yes
Systematic monitoring scheme: No
In-place land/water protection
Conservation sites identified: Yes, over entire range
Percentage of population protected by PAs: 91-100

Conservation Action in Place
Area based regional management plan: Yes
Occurs in at least one protected area: Yes
Invasive species control or prevention: Not Applicable
In-place species management
Harvest management plan: No
Successfully reintroduced or introduced benignly: Yes
In-place education
Subject to recent education and awareness programmes: Yes
Included in international legislation: Yes
Subject to any international management / trade controls: Yes

Conservation Actions Needed

(http://www.iucnredlist.org/technical-documents/classification-schemes)

Conservation Action Needed	Notes
1. Land/water protection -> 1.1. Site/area protection	Ensure full protection of leuser ecosystem and establish protection management of batang toru forests
1. Land/water protection -> 1.2. Resource & habitat protection	Ensure legitimate/legal oil palm concessions establish corridors and set aside key habitats
2. Land/water management -> 2.1. Site/area management	Improved management and law enforcement in all protected areas, non-protected forests and concessions
2. Land/water management -> 2.3. Habitat & natural process restoration	Habitats where concessions or roads are shown to be illegal and prosecutions taken place to be confiscated (or cancelled) and restored to former condition (tripa swamps, illegal roads, etc.)
3. Species management -> 3.2. Species recovery	Key corridors to be identified and established to maintain all populations above minimum genetically viable levels

Conservation Action Needed	Notes
3. Species management -> 3.3. Species re-introduction -> 3.3.1. Reintroduction	-
5. Law & policy -> 5.1. Legislation -> 5.1.2. National level	-
5. Law & policy -> 5.1. Legislation -> 5.1.3. Sub-national level	-
5. Law & policy -> 5.4. Compliance and enforcement -> 5.4.2. National level	-

Research Needed

(http://www.iucnredlist.org/technical-documents/classification-schemes)

Research Needed	Notes
1. Research -> 1.1. Taxonomy	Determination of status of batang toru orangutans (as new subspecies or species)
1. Research -> 1.2. Population size, distribution & trends	New surveys in habitats between leuser and batang toru forests
1. Research -> 1.5. Threats	Chain of custody studies to publicise companies still engaged in (directly or supplied by) destruction of the leuser ecosystem
2. Conservation Planning -> 2.2. Area-based Management Plan	Effective management plans needed for batang toru and leuser ecosystem, fully endorsed by national, provincial and district governments
3. Monitoring -> 3.1. Population trends	Always needed
3. Monitoring -> 3.4. Habitat trends	Always needed

Additional Data Fields

Distribution
Lower elevation limit (m): 0
Upper elevation limit (m): 1,500
Population
Continuing decline of mature individuals: Yes
Extreme fluctuations: No
Population severely fragmented: Unknown

Population
Continuing decline in subpopulations: Unknown
Extreme fluctuations in subpopulations: No
All individuals in one subpopulation: No

Habitats and Ecology
Generation Length (years): 25
Movement patterns: Not a Migrant

Amendment

Amendment reason: This second amended version of the 2017 assessment was crated to attach supplementary information. Previous corrected assessments were created to update the distribution map.

The IUCN Red List Partnership

The IUCN Red List of Threatened Species™ is produced and managed by the IUCN Global Species Programme, the IUCN Species Survival Commission (SSC) and The IUCN Red List Partnership.

The IUCN Red List Partners are: ABQ BioPark; Arizona State University; BirdLife International; Botanic Gardens Conservation International; Conservation International; Missouri Botanical Garden; NatureServe; Re:wild; Royal Botanic Gardens, Kew; Sapienza University of Rome; Senckenberg Society for Nature Research; Texas A&M University; and Zoological Society of London.